

YASH TARUN

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EDUCATION

Indian Institute of Technology (ISM) Dhanbad

Dec 2020 - May 2025

B.Tech (Double Major), Computer Science and Engineering

GPA: 8.13 / 10.0

Relevant coursework: Data Structures & Algorithms, Deep Learning, Advanced Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, Probability & Statistics.

EXPERIENCE

Barclays - Technology Analyst

Aug 2025 - Present

Knutsford, United Kingdom

Python, SPDW, LLMs (Claude), Celonis, FastAPI

- Designed and shipped an **AI incident-intelligence layer** on **ServiceNow NowAssist** using Barclays' **Claude LLMs**, automating triage, assignee recommendation, and resolution suggestions across enterprise incident workflows.
- Built **retrieval pipelines** over historical incident data (**SPDW** via **Celonis**) to auto-correct descriptions and recommend fixes, improving **triage accuracy by 30%** and reducing manual intervention.
- Engineered a **context-aware clustering and recommendation system** that grouped similar incidents and triggered automated notifications, cutting **mean time to resolution (MTTR) by 20-25%**.

Barclays - Technology Analyst Intern

Jun 2024 - Aug 2024

Knutsford, United Kingdom

Python, Deep Learning, SQL

- Developed **machine-learning models** to automate control and validation processes for the **Group Technology Service Management (GTSM)** team, reducing manual review time by **20%**.
- Improved model accuracy from **75% to 82%** by replacing random forests with **deep neural networks**, and validated the change with rollback plans before promoting models to production.

PROJECTS

Suno - On-Device Multilingual Voice-Note Digester

[Demo](#) · [GitHub](#)

Python, PyTorch, LoRA, Gemma-3-4B, llama.cpp, ONNX, sqlite-vec, Modal

- Built a fully **on-device** pipeline that transcribes, summarizes, translates, and extracts action items from **Hindi, Tamil, and Bengali** voice notes, with automatic language identification and **IndicConformer-600M** int8 ONNX speech-to-text.
- Fine-tuned **Gemma-3-4B with LoRA** (Q4 GGUF, llama.cpp, JSON-grammar-constrained decoding), raising action-item **F1 from 0.55 to 0.76** and confidence-tag accuracy to **0.90-1.00** on a held-out 68-note evaluation.
- Designed an **eval-first harness** (training and evaluation on **Modal**, per-slice scoring by language and phrasing) plus **sqlite-vec** semantic search over summaries and tasks; runs **about 25s/note, fully local** on an 8 GB laptop.

MemoryPlace - Benchmarking Platform for Agentic-Memory Architectures

[Demo](#) · [GitHub](#)

Python, FastAPI, Modal, PostgreSQL, Next.js, TypeScript, OpenAI

- Built an open platform to publish, evaluate, and compare **agentic-memory architectures** (mem0, Zep, and others) against a standardized **reset/ingest/retrieve** interface, cutting the effort to benchmark a new memory system from **weeks to minutes**.
- Ran every evaluation in an isolated, disposable **Modal container** that installs each architecture's own dependencies, keeping untrusted code away from the backend, database, and other runs for **reproducible, comparable** results.
- Scored memory quality on the **LoCoMo** and **LongMemEval** benchmarks (**F1, ROUGE-L, SBERT**) and enforced that leaderboard entries originate only from Hub runs with full provenance (architecture version, dataset, model, and config).

CompCode - Codebase QA & Agentic Editing System

[Demo](#) · [GitHub](#)

Python, LangGraph, tree-sitter, Qdrant, OpenAI, FastAPI, Next.js, Langfuse

- Built an **LLM system** to query and edit large codebases in natural language, extracting function-level chunks with **tree-sitter AST** across **7 languages** and embedding them in a **Qdrant** vector database.
- Designed a **hybrid retrieval pipeline** combining vector similarity with **call-graph expansion** and context compression, cutting irrelevant context retrieval by **about 30%** versus pure semantic search.
- Implemented a **LangGraph multi-agent workflow** (route, retrieve, plan, human-approved apply, verify) that proposes edits as **reviewable diffs**, self-corrects on compile/lint/test failures, and traces every step to **Langfuse**.

TECHNICAL SKILLS

Languages

Python, C++, Java, C, SQL, JavaScript / TypeScript

ML & AI

Deep Learning, PyTorch, NLP, LLM fine-tuning (LoRA), model quantization (GGUF / ONNX), retrieval-augmented generation (RAG), vector databases (Qdrant, sqlite-vec), prompt engineering, LangGraph, Hugging Face, model evaluation, Modal

Tools & Systems

FastAPI, Node.js, Next.js, REST APIs, Docker, Git, Linux, PostgreSQL, MongoDB, Redis

CS Foundations

Data Structures & Algorithms, System Design, OOP, Operating Systems, DBMS, Computer Networks

ACHIEVEMENTS

- Codeforces Specialist** (max rating 1590); global rank under 100 in Round 970.
- CodeChef 4-star (1892), LeetCode 1686; **800+** problems solved across platforms.
- Runner-up (2nd place)**, **Hackfest 2022**, IIT (ISM) Dhanbad, organized by Atlassian.
- Top 1 percentile** in JEE Mains and JEE Advanced 2020.
- Event Co-Head, Concetto'22, the annual techno-management fest of IIT (ISM) Dhanbad.